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Nonlinear programming reformulation of dynamic flux balance analysis models

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ABSTRACT

Dynamic flux balance analysis (dFBA) models are very popular in systems biology because of the many possible applications. dFBA consists of dynamic mass balance equations that describe the concentration of external metabolites and an optimization model to compute the internal flux distribution of the cell. Here, we applied a nonlinear programming formulation for solving dFBA models. The ODE system is discretized using the orthogonal collocation technique and the optimization problem is replaced by its first-order optimality conditions. In addition, an adaptive mesh scheme is applied to improve performance by efficiently handling changes in the active set of constraints. This formulation ensures total differentiability of the model, which can be solved by large-scale solvers and take advantage of automatic differentiation packages. The new approach presents results equivalent to state-of-the-art methods and outperforms them in dynamic optimization problems by avoiding calling an external solver to handle the embedded optimization model.

1. Introduction

The description of the dynamics of metabolism is relevant for a range of research fields such as systems biology, synthetic biology, metabolic engineering, and bioprocess engineering (Nielsen, 2017). Dynamic models have been used for a variety of applications such as predicting the effect of gene deletions on bioprocess productivity (Hjersted et al., 2007), optimal control of bioprocesses (Chang et al., 2016), and understating the effect of diseases on human cell metabolism (Ben Guebila and Thiele, 2021). A widely applied dynamic model of metabolism is the so-called Dynamic Flux Balance Analysis (dFBA) (Mahadevan et al., 2002). The main hypothesis in dFBA models is that the intracellular dynamics are faster than the extracellular dynamics, and hence, the intracellular metabolites can be described by a steady-state model. In order to implement this method, a genome-scale metabolic network is used. The networks are built using omics information, mapping transformations between substrates and products catalyzed by enzymes (Thiele and Palsson, 2010). The metabolic networks are expressed in the form of a matrix (stoichiometric matrix), where each column corresponds to an enzymatic reaction and each row to a metabolite. Because typically these matrices have more fluxes than metabolites, an undetermined system of equations is to be solved. An optimization problem is formulated for that purpose using a biological meaningful objective function. This process is called Flux Balance Analysis (FBA) (Orth et al., 2010). The dynamic version of FBA is implemented by applying mass balance equations to describe the behavior of the extracellular metabolites, in conjunction with kinetic expressions aimed to describe the uptake of substrates. The solution of dFBA models presents some challenges that are summarized below:

- Stiff behavior: Normally dFBA models present stiff behavior, especially in situations as diauxic growth (Mahadevan et al., 2002; Hjersted et al., 2007).
- Non-unique solutions: in most cases, the solution of the optimization problem (FBA) is not unique. In consequence, situations where the fluxes “jump” between different optimal solutions can arise.
- Feasibility: When the optimization solver cannot find a feasible solution for the FBA problem, the interaction with an ODE solver can lead to errors.
- Scalability: The size of the metabolic networks applied in FBA can be large, therefore methods should scale adequately for those situations.
- Nonlinearity: The presence of nonlinear constraints or objective functions can substantially increase the computational demand.
- Differentiability: When the dFBA models are used inside an optimal control or parameter estimation architecture, the first and second derivatives of the model must be computed.

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The simplest method for solving a dFBA model is the Static optimization approach (SOA) (Mahadevan et al., 2002). In the SOA the ODE system is discretized using an explicit Euler method and the FBA is solved at each time step. For stiff problems, the Euler method would need a small step, then the FBA needs to be solved a large number of times. Another usual approach is the direct approach (DA) (Zhuang et al., 2011), where an adaptive size solver for the ODE system is used and the number of optimizations decreases in comparison with the SOA method. Both SOA and DA are easy to implement but the solution of the optimization problem can make the simulations fail.

Modifications to the DA have been proposed in order to address this issue, like event-based methods. Gomez et al. (2014) developed an event-based methodology that detects changes in active-set of the FBA model (Harwood et al., 2016). This makes the solution faster because the FBA problem does not need to be solved at each time step. They also applied a lexicographic optimization methodology to deal with the non-unique solutions of FBA, wherein the first optimization problem solves a feasibility problem. Other event-based methods that compute an optimal basis for the FBA feasible space have also been proposed (Ploch et al., 2020; Brunner and Chia, 2020). The drawback of these formulations is the need for continuous monitoring the active-set of the optimization problem that can increase with the size of the metabolic network. Also, the model is non-differentiable at the points of change of the active-set.

Another class of approaches proposes the reformulation of the dFBA using the Karush–Kuhn–Tucker (KKT) conditions of the FBA problem. Ploch et al. (2020) reformulated the dFBA model as a nonsmooth DAE system and applied a homotopy continuation to avoid the fluxes from jumping between optimal solutions. Scott et al. (2018) used an interior point reformulation of FBA. The solvers failed to solve the generated DAE system and they needed to apply DAE index reduction to obtain an implicit ODE system. Dynamic optimization problems using genome-scale networks were solved and the model derivative information was computed solving the sensitivity equations. Surrogate models of dFBA have also been proposed, they focused mainly on Model Predictive Control (MPC) applications, for introducing robustness using a polynomial chaos expansion (Kumar and Budman, 2017) or to allow the resolution of large genome-scale models in a short time and high convergence rate (Oliveira et al., 2021).

Finally, there is a class of approaches that reformulate the problem as a Nonlinear Programming (NLP) problem. These approaches discretize the ODE system using a technique such as orthogonal collocation on finite elements, which are included as constraints in the optimization problem. The simplest version is the Dynamic Optimization approach (DOA) (Mahadevan et al., 2002) that uses some instantaneous objective function such as the maximization of biomass formation. Raghunathan et al. (2003) formulate a parameter estimation problem for dFBA problems using the KKT conditions for the FBA problem. Chang et al. (2016) did the same for MPC formulation. The key advantage of these NLP formulations is that they can easily incorporate path constraints, nonlinear constraints/objective functions and they can be easily integrated into parameter estimation and optimal control problems as a single-level problem. Also, the FBA optimization does not need to be solved a large number of times during the simulation. The main drawback of this strategy is the need to solve large NLP that can be difficult to initialize and can suffer from convergence problems (Mahadevan et al., 2002; Chang et al., 2016).

Here, we applied a well-known NLP reformulation for the dFBA models that is fast, precise, and can be applied to large genome-scale networks. The methodology consists of incorporating a KKT reformulation of a parsimonious FBA problem and applying a collocation technique to the ODE system. In addition, we demonstrated by case studies when this approach can fail and how overcoming that using an adaptive mesh scheme and imposing constraints on the fluxes variables within the finite elements. The method was implemented using the free open source and high-performance language Julia (Bezanson et al., 2017). Furthermore, the utilization of automatic differentiation code packages can allow the fast computation of first and second-order derivatives. This represents a considerable advance in comparison to the simulation-based approaches that need to compute the derivatives by finite differences, sensitivities equations, or applying derivative-free methods. Also, the application of a large-scale NLP solver (e.g. IPOPT) can considerably reduce the computational time and handle large optimization problems by exploring the sparsity of the NLP systems (Shin et al., 2019). We named this approach to solve dFBA models Direct Collocation dFBA (DC dFBA).

This paper is organized as follows: Section 2 presents the dFBA mathematical formalism and the DC dFBA formulation and Section 3 presents the case studies that were solved using the DC dFBA.

2. Problem statement

In this section, first the mathematical formulation of the dFBA models is introduced. Next, the transformations used to obtain the NLP formulation are presented. First, the discretization of the ODE system using the orthogonal collocation technique is introduced followed by the KKT transformation of the pFBA problem. Finally, the DC dFBA approach is presented.

2.1. Flux Balance Analysis (FBA)

Metabolic networks of a variety of microorganisms have been reconstructed so far (Palsson, 2015). These networks describe the metabolic transformations inside the cell, from the substrates to the internal metabolites, biomass and, bioproducts. The utilization of these networks can significantly improve the prediction capabilities of mathematical models in comparison with unstructured models (e.g. Monod model) (Hodgson et al., 2004). The information contained in a metabolic network can be synthesized in a matrix format, the stoichiometric matrix (S). The S rows correspond to metabolites and the columns to enzymatic reactions. Typically, more reactions than metabolites are present on these networks (Palsson, 2015). All possible fluxes distributions for the steady-state cell metabolism are contained in the null space of the stoichiometric matrix ($\boldsymbol{Sv} = 0$). Therefore, in order to find a solution in that space, an undetermined system must be solved.

Flux Balance Analysis (FBA) assumes that the cell has a biological objective function, typically maximizing biomass formation (Orth et al., 2010). An optimization problem is built in order to obtain a flux distribution. The FBA can be formulated as a Linear Programming (LP) problem as follows:

$$\begin{aligned}
&\max_{\boldsymbol{v}} \quad c^T \boldsymbol{v} \\
&\text{s.t.} \\
&\boldsymbol{S}\boldsymbol{v} = 0 \\
&\boldsymbol{v}_{\min} \leq \boldsymbol{v} \leq \boldsymbol{v}_{\max}
\end{aligned} \tag{1}$$

where $\boldsymbol{v}$ are the metabolic fluxes; c^T a vector that combines the fluxes to form the objective function; S is the stoichiometric matrix; $\boldsymbol{v}_{\min}$ and $\boldsymbol{v}_{\max}$ are the upper bound and lower bound of the fluxes respectively. In summary, this optimization problem's decision variables are the fluxes in the metabolic network. While the equality constraints are the metabolic network steady-state solutions, and the inequality constraints are applied to specific metabolic fluxes in the network, usually limiting some substrate uptake. Typically, FBA problems have multiple optimal solutions (Orth et al., 2010; Palsson, 2015), though for networks with enough constraints, there can be unique solutions. However, this is not the case for most genome-scale networks where there are not enough constraints (Mahadevan and Schilling, 2003). In order to deal with the multiplicity of solutions, some methodologies have been proposed in the literature. Gomez et al. (2014) applied a lexicographic optimization using a sequence of cells objective functions. The main challenge of the approach is to define this sequence adequately. Scott et al. (2018) applied an interior-point formulation of the KKT conditions of the FBA problem to guarantee a unique arbitrary solution. Another formulation is the so-called parsimonious FBA (pFBA) (Lewis et al., 2010). pFBA assumes that the cell will utilize the pathways that have minimal overall intracellular flux, which would be associated with a small cost of enzyme synthesis. pFBA has presented good agreement with experimental data (Machado and Herrgård, 2014). The pFBA consists in solving two sequential linear optimization problems. First, the standard FBA (Eq. (1)) is solved to obtain the maximum objective value ($c^T\boldsymbol{v}$), and then a second LP is solved minimizing the l_1 norm of the flux vector with an additional constraint coming from the previous FBA problem ($c^T\boldsymbol{v} = \max c^T\boldsymbol{v}$).

$$\begin{aligned}
&\min_{\boldsymbol{v}} \quad \sum v \\
&\text{s.t.} \\
&c^T\boldsymbol{v} = \max c^T\boldsymbol{v} \\
&\boldsymbol{S}\boldsymbol{v} = 0 \\
&\boldsymbol{v}_{\min} \le \boldsymbol{v} \le \boldsymbol{v}_{\max}
\end{aligned} \tag{2}$$

2.2. Dynamic Flux Balance Analysis (dFBA)

The dynamic version of the FBA model, the dFBA, is formulated assuming that the intracellular dynamic metabolism is by far much faster than the extracellular dynamics. Thus, the internal metabolism can be represented by the steady-state FBA model (optimization problem), while the extracellular metabolites are modeled by mass balance equations taking into account the dynamics (Ordinary Differential Equation (ODE) system). In addition, kinetic expressions are used to model the uptake of substrates. The ODE system that describes the extracellular dynamics is parameterized by the flux vector (v) that is obtained from solving the FBA optimization problem. Therefore, the dFBA can be formulated as an ordinary differential equation system with embedded optimization (ODEO) problem:

$$\begin{aligned}
\frac{dx}{dt} &= F(x, v) \\
v_{uptake} &= g(x) \\
x &\ge 0 \\
v &= FBA(v_{uptake}) \quad \text{(Eq. (1))}
\end{aligned} \tag{3}$$

where x are the concentrations of the external metabolites; F is the right-hand side of the mass balance equations; g is the kinetic expression to calculate the uptake fluxes v_{uptake}. The dFBA model is an alternative to the utilization of kinetic models of metabolism that need the knowledge of enzymatic reactions mechanism and kinetic parameters from the internal reactions. The dFBA only uses a few kinetic parameters on the uptake reactions that are considered limiting steps. However, despite their simplicity, the dFBA models are challenging ODEO problems to solve. In the next sections, an NLP transformation of the ODEO system will be elaborated.

2.3. ODE discretization using orthogonal collocation on finite elements

The ODE system of the dFBA model is based on the external metabolite's mass balances. Efficient ODE solvers are available for stiff problems like dFBA models, and many of them have been applied until now (Gomez et al., 2014; Scott et al., 2018). However, the interaction of the optimization solver with the FBA problem can lead to errors, especially if the optimization solver does not return any solution (infeasible problem) at a given integration step. Another difficulty is that the optimization problem needs to be solved many times, at each time step of the ODE solver. For large metabolic networks or more computationally demanding versions of FBA like pFBA (Eq. (2)), the sequence of optimizations can get computationally cumbersome.

Alternatively, instead of using an ODE solver, the ODE system can be discretized using the orthogonal collocation technique (Biegler, 2010) (also known as direct transcription). This technique was demonstrated to be equivalent to an implicit Runge–Kutta method with high-order accuracy and strong stability properties (Biegler, 2010). Here, we discretized the ODE system using orthogonal collocation on three Radau collocation points. Also, the time domain was divided into finite elements.

2.4. Karush–Kuhn–Tucker (KKT) reformulation of FBA

While the ODE system can be transformed into an algebraic equation system using orthogonal collocation, the FBA optimization problem can be transformed into a system of algebraic equations. This can be done by applying either the duality theory (Maranas and Zomorrodi, 2016) or the first-order necessary optimality condition (Karush–Kuhn–Tucker-KKT) (Raghunathan et al., 2003). While the duality theory transformation avoids the complementarity constraints on the model, this formulation leads to a nonconvex NLP, for which only locally optimal solutions can be guaranteed (Raghunathan et al., 2003). For this reason we decided to apply the KKT conditions here.

Typically, the FBA has multiple optimal solutions, Gomez et al. (2014) applied lexicographic optimization to deal with the multiplicity of solutions, however, the number of optimization problems increases, and the order of objective functions must be supplied. Scott et al. (2018) applied an interior-point method to obtain an arbitrary unique solution in the interior of the feasible space. pFBA (Eq. (2)) has shown a good agreement with experimental data (Lewis et al., 2010; Machado and Herrgård, 2014), therefore, here an alternative formulation for the pFBA problem presented by Ploch et al. (2020) was implemented:

$$\begin{aligned}
&\max_{\boldsymbol{v}} \quad c^T\boldsymbol{v} - v^T W v \\
&\text{s.t.} \\
&\boldsymbol{S}\boldsymbol{v} = 0 \\
&\boldsymbol{v}_{\min} \le \boldsymbol{v} \le \boldsymbol{v}_{\max}
\end{aligned} \tag{4}$$

The Eq. (4) consists of the FBA formulation with the addition of a quadratic regularization term to search for solutions with minimal overall intracellular flux. As observed by Ploch et al. (2020), for a small value of parameter w the solution of Eq. (4) is also a solution of Eq. (1). The LP FBA is then transformed into a quadratic programming (QP) problem that is also convex (Ploch et al., 2020). The KKT conditions for Eq. (4) are:

$$\begin{aligned}
&c - W v + S^T\lambda + \alpha^U + \alpha^L = 0 \\
&(v - \boldsymbol{v}_{\max})\alpha^U = 0 \\
&(\boldsymbol{v}_{\min} - v)\alpha^L = 0 \\
&\alpha^U, \alpha^L \ge 0
\end{aligned} \tag{5}$$

where λ, α^U, and α^L are the multipliers corresponding to the stoichiometric equation, fluxes lower bound and fluxes upper bound, respectively. W is the positive definite and diagonal regularization matrix with small positive values on the diagonal. Besides the KKT conditions, a constraint qualification must hold at the optimum point. The Linear independence constraint qualification (LICQ) is the most frequently used, which requires the active constraints' gradients to be linearly independent (Biegler, 2010).

The Eq. (5) belongs to a class of problems called Mathematical Programs with Complementarity Constraints (MPCC) (Baumrucker et al., 2008). Complementarity constraints are related constraints that at least one of them must be active. This MPCC is particularly changeling to solve because they violate the LICQ (Baumrucker et al., 2008). A series of regularization methods have been proposed to deal with the MPCC. Some methods consist of just relaxing the right-hand side of the complementarity constraints using a small value and solving a series of simulations decreasing this value (Baumrucker et al., 2008). Applying an extreme-ray-based transformation (Zhao et al., 2017), or an interior-point transformation (Scott et al., 2018) have also been proposed. Also, a Mixed-Integer Nonlinear Programming (MINLP) transformation can be applied (Maranas and Zomorrodi, 2016).

Finally, a penalization method can be applied when the complementarity constraints are inserted in the objective function as a penalty parameter. Here, we decided to apply the penalty method to deal with the MPCC because it can be solved in a single optimization problem and it has also presented good results with NLP interior-point solvers (Baumrucker et al., 2008), which was applied in this work.

Remark. It is essential to point out that even though the penalty method used to deal with complementary constraints can overcome the LICQ violations associated with this class of problems (Baumrucker et al., 2008), it is not possible to guarantee that the LICQ will hold in all cases (Zhao et al., 2017). In this work, we assume that LICQ hold, but a complete analysis of this hypothesis must be carried out in future works.

2.5. DC dFBA implementation

We are now able to build an NLP formulation of the dFBA model by applying the orthogonal collocation on finite elements and the KKT conditions. The l_1 penalization term with the complementarity constraints was inserted on the objective function (Baumrucker et al., 2008) as follows:

$$
\begin{aligned}
&\max_{x,v} \quad \Phi(x,v) - \rho((v - \boldsymbol{v}_{\max})\alpha^U + (\boldsymbol{v}_{\min} - v)\alpha^L) \\
&\text{s.t.} \\
&\frac{dx}{dt} = F(x,v) \\
&x \geq 0 \\
&v_{uptake} = g(x) \\
&lb \leq v \leq ub \\
&c - W\,v + S^T\lambda + \alpha^U + \alpha^L = 0 \\
&\alpha^U, \alpha^L \geq 0
\end{aligned}
\tag{6}
$$

where ρ is the penalization parameter and $\Phi(x,v)$ is the objective function. For the case of a simple dFBA simulation, the value of $\Phi(x,v)$ is zero. However, $\Phi(x,v)$ can also be a sum of squares of residues for a parameter estimation problem, or an optimal control objective. That is one advantage of the NLP formulation, which can be easily adapted for different applications of the dFBA model. Moreover, the NLP formulation allows direct gradient and Hessian evaluations, in contrast to the sensitivity calculations when DAE solvers are used, which can cause convergence difficulties (Biegler, 2010) and require that the dynamic model be simulated many times (Shin et al., 2019).

On the other hand, efficient large-scale NLP solvers are required for efficient solution of the NLP formulation. Therefore, we implemented the DC dFBA formulation on JULIA language (Bezanson et al., 2017) with JUMP as mathematical programming language (Dunning et al., 2017). The NLP was solved using the IPOPT solver (Wächter and Biegler, 2006) with the linear solver MA27 and using the automatic differentiation package (Rackauckas and Nie, 2017) for fast computation of the gradient and Hessian. An overview of the DC dFBA approach can be found in Fig. 1. All calculations were performed on a laptop equipped with an Intel Core i5-1135G7 CPU, 2.40 GHz and 16 GB RAM.

3. Results and discussion

A series of case studies are presented in this section for different applications of dFBA models. First, the DC dFBA is applied to the simple *E. coli* core model. The need for flux constraints and an adaptive mesh scheme is justified based on diauxic growth simulations. In the second case study, the *E. coli* iJR904 model was applied, where the effect of the metabolic network size is explored. In the next case study the *E. coli* iJO1366 model was applied to estimate the optimal profile of genetic alterations for D-lactic acid production. Case study four evaluates the application of DC dFBA for *S. cerevisiae* iND750 model for the dynamic optimization of a bioreactor. Finally, the DC dFBA is applied to *S. cerevisiae* Yeast 8.3 model, which is the biggest metabolic network applied in this work, solves a parameter estimation problem.

3.1. Case study 1: Escherichia coli core model

The *E. coli* core model was chosen to be the first case study to test the methodology proposed because it is a simple and well-established model. First, a simulation of the aerobic growth on a medium containing glucose and acetate was performed. The model consists of mass balances for glucose (G), acetate (A), and biomass (X). Also, a Michaelis–Menten equation was applied to describing the glucose uptake:

$$
\begin{aligned}
&\frac{dX}{dt} = \mu X \\
&\frac{dG}{dt} = -v_g X \\
&\frac{dA}{dt} = v_A X \\
&v_g \leq v_{g,max}\frac{G}{K_g + G} \\
&v_o \leq v_{o,max} \\
&X, G, A \geq 0 \\
&v_A,\ \mu = pFBA(v_o, v_g)
\end{aligned}
\tag{7}
$$

where μ, v_g, v_o, v_A are the growth rate, the glucose uptake flux, the oxygen uptake flux and the acetate flux, respectively. $K_g = 0.01$ mmol is the affinity constant for the substrate. $v_{g,max} = 10.5$ mmol/gdw h and $v_{o,max} = 19.0$ mmol/gdw h are the maximum uptake rate for glucose and oxygen, respectively. The acetate can be produced during aerobic growth on glucose or consumed when glucose is exhausted. It was assumed that the acetate can freely diffuse through the cell, and the acetate lower bound flux was set to -2.5 mmol/gdw h.

The first simulation was performed until the moment of glucose exhaustion, for that case, no acetate consumption should occur. The DC dFBA formulation presented in Eq. (6) was used with a equidistant mesh of six elements and $\Phi(x,v) = 0$. In order to validate the solution obtained by the NLP formulation, the DFBAlab simulator was used as Ref. Gomez et al. (2014). The lexicographic optimization order in DFBAlab was maximizing growth rate, minimizing glucose uptake, and maximizing acetate production. Also, the DA approach was applied using an explicit Runge–Kutta method from JULIA's DifferentialEquations package. The results showed a good agreement between the different approaches (Fig. 2A).

3.1.1. E. coli diauxic growth

A classic problem that can be simulated by dFBA models is the diauxic growth phenomena, when a microorganism in the presence of two substrates presents a two-phase growth, one for each substrate (Mahadevan et al., 2002). The dFBA should predict the right point when the preferential substrate is exhausted, and the cells start to consume the second substrate. In this example, *E. coli* first consumes the glucose and then acetate (Mahadevan et al., 2002). The DC dFBA approach described the system dynamics poorly for this problem (Fig. 2b). In order to diagnose the reasons for the failure of the DC dFBA approach to converge to the right solution, the acetate flux (v_{ac}) profile and the Lagrange multipliers associated to the acetate flux lower bound (α^L_{ac}) were examined more carefully. Table 1 shows the acetate flux and multipliers for each node in each finite element. As can be seen, the flux of acetate alternates between the two possible optimal solutions, one consuming glucose and another producing acetate. The last finite element is placed in the middle of the transition between the two

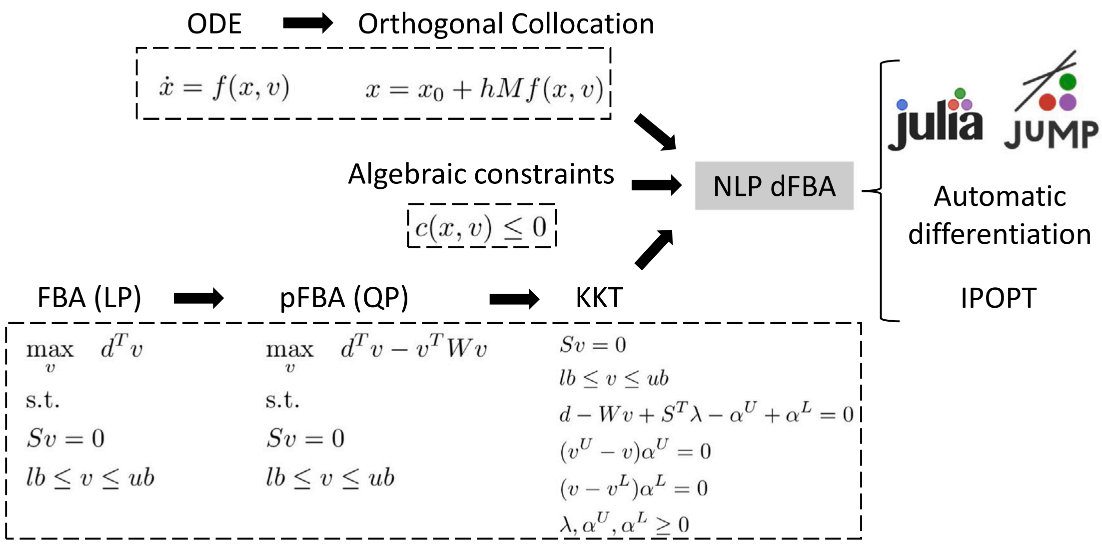

Fig. 1. Overview of the DC dFBA approach. The ODE system was discretized using the orthogonal collocation technique on finite elements and inserted on the DC dFBA as constraint. The algebraic constraints of dFBA related to uptake kinetics were directly inserted on the optimization problem as constraints. The FBA problem was first transformed into a pFBA by the addition of a quadratic regularization term. Then the KKT conditions of the pFBA were computed and inserted on the DC dFBA as constraints. Finally the DC dFBA was implemented in Julia language using the JUMP environment (Dunning et al., 2017). The automatic differentiation package was used to compute the derivatives and the solver IPOPT to solve the resulting NLP problem.

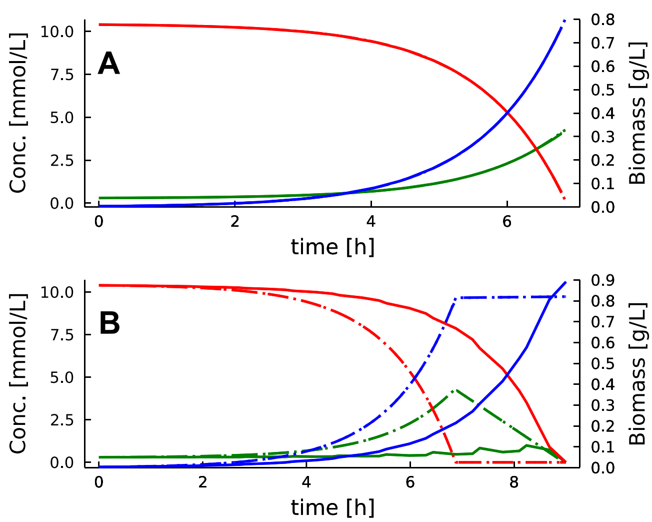

Fig. 2. *Escherichia coli* core model simulation until (a) glucose exhaustion and (b) acetate exhaustion. The concentration profiles are glucose (red), biomass (blue) and acetate (green). DC dFBA (solid), DFBAlab (dotted) and direct approach (dashed). All the methods presented the same result for the *E. coli* growth on glucose (a), however, the DC dFBA failed to describe the diauxic growth (b).

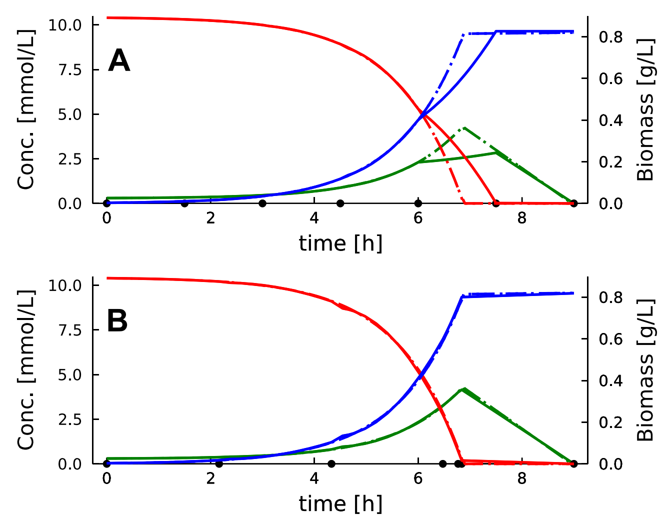

Fig. 3. *E. coli* core model diauxic growth simulation (a) imposing only flux constraint and (b) imposing flux constraint and an adaptive mesh on the DC dFBA. The concentration profiles are glucose (red), biomass (blue) and acetate (green). DC dFBA (solid), DFBAlab (dotted) and the direct approach (dashed). RMSE [%] for glucose, acetate and biomass for DFBAlab/DA were 0.81/1.57, 0.70/1.07, 0.89/1.65, respectively. CPU time for DC dFBA, DFBAlab and DA were 0.46, 0.99, 13.79 (seconds), respectively. The position of the finite elements is indicated with black circles.

phases. The multipliers for the last element are equal to zero for the first two nodes, and different from zero in the last one. This indicates that the acetate lower bound flux constraint becomes active within the element.

Biegler et al. (2002) noted that when there is a change of the active-set within a finite element, convergence problems can arise. They proposed an *ad hoc* method to deal with this problem. First, the NLP problem is solved and then it is evaluated if there is a change of active set within some finite element. Then, for these elements, the size is set as a variable. The methodology presented good results for optimal control applications, however, the need to solve the problem more than once makes the method not so practical to apply. Here, in order to deal with this problem, an extra constraint was imposed into Eq. (6) to prevent that flux distribution within a finite element from having a considerable change:

$$v_i^k - v_i^{k-1} \leq \epsilon \qquad k = 1, \ldots, ncp \tag{8}$$

where ncp is the number of collocation points, and ϵ is the allowed change in each flux. When this new constraint is added to the solution of the *E. coli* diauxic growth problem (using $\epsilon = 0$), the profiles that are obtained are much closer to those from the other approaches (Fig. 3a). Looking again at the acetate flux and multipliers profiles on the finite elements (Table 1), it is possible to see that there are no changes within a finite element. Moreover, only the last element is in the phase of consuming acetate.

Table 1
Acetate flux and acetate lower bound multiplier profiles for each node in each finite element for the solution of *E. coli* diauxic growth using the DC dFBA. When no flux constraint imposes the solution to jump between the two local solutions. However, when the flux constraints were imposed, there is no change in the flux and multipliers within an element.

	Element	1			2			3		
	Node	1	2	3	1	2	3	1	2	3
Without flux const.	α^L_{ac}	9.86e−9	9.86e−9	9.86e−9	9.86e−9	9.86e−9	9.86e−9	9.86e−9	9.86e−9	9.86e−9
	v_{ac}	4.13	−2.50	4.13	4.13	−2.50	4.13	4.13	−2.50	4.12
Applying flux const.	α^L_{ac}	9.62e−9	9.62e−9	9.62e−9	9.62e−9	9.62e−9	9.62e−9	9.62e−9	9.62e−9	9.62e−9
	v_{ac}	4.13	4.13	4.13	4.13	4.13	4.13	4.04	4.04	4.04
	Element	4			5			6		
	Node	1	2	3	1	2	3	1	2	3
Without flux const.	α^L_{ac}	9.86e−9	9.86e−9	9.86e−9	9.86e−9	9.86e−9	9.86e−9	9.86e−9	−0.02	−0.02
	v_{ac}	4.12	−2.50	4.12	4.12	−2.50	4.08	3.76	−2.50	−2.5
Applying flux const.	α^L_{ac}	9.59e−9	9.59e−9	9.59e−9	9.51e−9	9.51e−9	9.51e−9	−0.02	−0.02	−0.02
	v_{ac}	3.73	3.73	3.73	2.62	2.62	2.62	−2.50	−2.50	−2.50

3.1.2. Adaptive mesh

Despite the profiles presenting a better agreement with other approaches when the flux constraints were imposed (Fig. 3b), the transition between the two phases is still poorly described. This takes place because the last element, which is in the consuming acetate phase, starts before the transition of the two phases. Therefore, to solve this issue, a simple adaptive mesh strategy was implemented as described in (Biegler, 2010):

$$\sum_{k=1}^{ne} h_k = t_{end}, \quad h_k \in [(1-\gamma)\overline{h}, (1+\gamma)\overline{h}] \tag{9}$$

In this strategy, the size of each element (h_k) is now a variable in the NLP and it is allowed to change in a range based on the equidistant element size ($\overline{h}$). For this problem, the element sizes were allowed to change freely ($\gamma = 1$). The profiles now have a good agreement with other approaches (Fig. 3b). The position of the elements is indicated with circles on the bottom of Fig. 3. The last element now begins exactly at the point of transition between phases, then the profile can be more accurately described. In order to have a more precise measure of the difference between the approaches to solve the dFBA model, the Normalized Root Mean Square Error (RMSE) was calculated as follows:

$$RMSE\ [\%] = \frac{\sqrt{\frac{\sum_{i=1}^{N}(x_i^a - x_i^b)^2}{N}}}{max(x_i)} \tag{10}$$

where N is the number of samples, the indexes a and b correspond to different approaches and $max(x_i)$ is the maximum value of the metabolite. RMSE [%] for glucose, acetate and biomass for DFBAlab/DA were 0.81/1.57, 0.70/1.07, 0.89/1.65, respectively. CPU times for DC dFBA, DFBAlab and DA were 0.46, 0.99, 13.79 (s), respectively. The adaptive mesh makes the problem more flexible with the cost of a bigger optimization problem to solve. However, by tuning the γ parameter this issue can be softened. For this small metabolic network, no considerable difference in the time to solve the problem was noted between DC dFBA or DFBAlab. However, the DA method spends much more time solving the problem. This can be explained by the need for the DA to solve more optimization problems. Two for each step, because the bi-level version of pFBA was used.

3.2. Case study 2: E. coli iJR904 model

The size of the metabolic network can have a considerable impact on the time and precision of dFBA simulations. Many authors suggested that an optimization approach would be intractable for genome-scale metabolic networks simulations (Gomez et al., 2014; Scott et al., 2018; Oliveira et al., 2021). Indeed, optimization approaches can suffer from scalability issues if the optimization problem becomes too big to be handled. Therefore, the *E. coli* iJR904 genome-scale model (GSM) (Reed et al., 2003) was applied to simulate the diauxic growth problem in Eq. (7). The *E. coli* iJR904 model has 761 metabolites and 1075 reactions, which is ten times bigger than the *E. coli* core model solved in the last section. The problem was solved using 9 adaptive elements. The results are shown in Fig. 4, the RMSE [%] for glucose, acetate, and biomass for DFBAlab/DA were 0.85/4.88, 4.17/12.7, 0.79/4.48, respectively. CPU times for DC dFBA, DFBAlab, and DA were 101.11, 1.05, and 204.62 s, respectively.

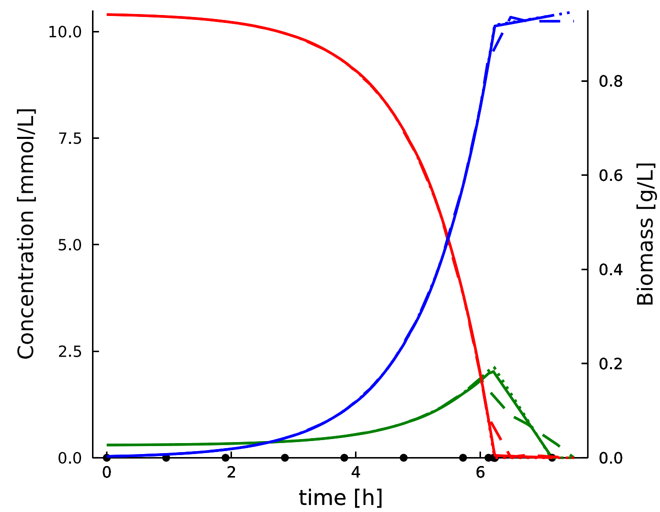

Fig. 4. *E. coli* iJR904 model diauxic growth simulation. The concentration profiles are glucose (red), biomass (blue) and acetate (green). DC dFBA (solid), DFBAlab (dotted) and direct approach (dashed). RMSE [%] for glucose, acetate and biomass for DFBAlab/DA are 0.85/4.88, 4.17/12.7, 0.79/4.48, respectively. CPU seconds for DC dFBA, DFBAlab and DA were 101.11, 1.05 and 204.62, respectively. The position of the finite elements is indicated with black circles.

The DA method failed to predict the transition between the two substrate phases, thus explaining the higher RMSE deviation. Indeed, the DC dFBA had a significant increment in computational time for the GSM while the DFBAlab had almost the same CPU times. This can be explained by the fact that it is much easier to solve bigger LP problems than NLP problems. Moreover, the fact that the DFBAlab computes an optimal basis for the FBA problem and avoids solving the optimization problem many times during the simulations explains the differences for the DA approach. It is important to highlight that despite the bigger amount of time to solve the DC dFBA, it is still a short time and also, the utilization of initial guess for the NLP problem can reduce that time (Shin et al., 2019).

3.2.1. Dynamic control of metabolism

A useful application of dFBA model is the dynamic control of metabolism. Gadkar et al. (2005) demonstrated that it is possible to

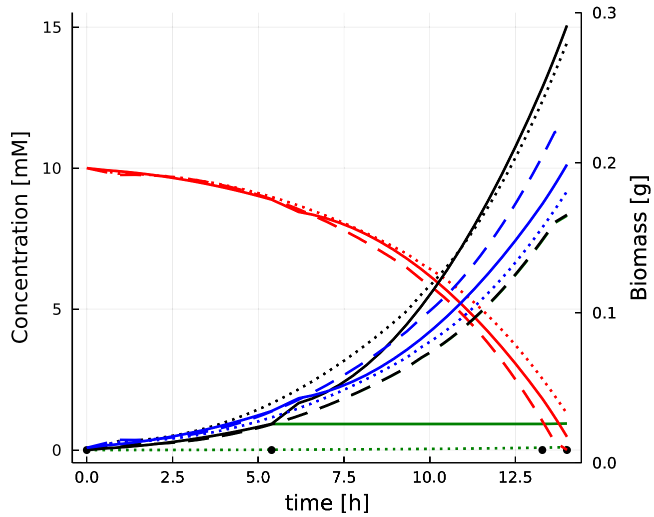

Fig. 5. Dynamic control of metabolism applying the *E. coli* iJR904 GSM solving by the DC dFBA and DFBAlab approaches with identical results (solid). Static deletion simulation (dotted) and wild-type simulation (dashed) are also presented. The concentration profiles are glucose (red), biomass (blue), acetate (green) and ethanol (black). The position of the finite elements is indicated with black circles.

apply dFBA models to estimate the optimal profile of metabolic fluxes to maximize the productivity of a bioproduct. The authors study the problem of maximizing ethanol production by *E. coli* regulating the flux through the acetate pathway. When acetate is being produced the cell can produce more ATP and have a high growth rate, on the other hand, when no acetate is being produced, the growth rate decreases but the ethanol production increases. The role of the optimization problem is to find the right time to switch from one state to another. The problem was formulated as a bi-level optimization problem, in the outer level they maximize ethanol at the end of the batch by manipulating the time of regulation (t_{reg}). At the inner level, the dFBA model was solved using the direct approach. Here, the same problem was solved using the *E. coli* iJR904 metabolic network. The problem was solved by the DC dFBA in a single-level optimization and the objective function in Eq. (6) was set to $\Phi(x, v) = ethanol(end)$. Three finite elements were used, in the last two elements the constraint of a null flux on the acetate pathway was imposed. Because the size of the elements can change, the NLP can choose the t_{reg} based on the position of these elements.

Fig. 5 shows the dynamic flux profile computed by DC dFBA and the profiles for the wild-type strain and the static deletion strategy (when the acetate pathway has been blocked since the beginning of the batch). As expected, the dynamic control strategy was able to increase ethanol production by manipulating the flux through acetate. Even though the direct comparison with the results presented in Gadkar et al. (2005) is not possible because here a GSM model was applied instead of a core model, the profiles are very similar.

The problem was also solved in DFBAlab as a bi-level optimization problem using the fmincon (sqp method) solver in MATLAB. In the outer optimization problem, the t_{reg} was set as a variable, and in the inner problem, two DFBAlab simulations were performed (one for each phenotype) based on the t_{reg}. The formulation of this problem in DFBAlab illustrates that the implementation is not so straightforward as the dFBA is already formulated as an optimization problem. The results of both methods were similar (Table 2) in terms of the time of switch and the CPUs time to solve the problem. Despite the fact that dFBAlab presented a much shorter time of simulation for the diauxic growth on *E. coli*, here, because the DC dFBA is already formulated as an optimization problem, the computational cost to perform a simulation is almost the same as performing an optimal control simulation. This difference will be more relevant when solving larger problems as will be evident in the next case studies.

Table 2
Comparison of DC dFBA and DFBAlab for solving the dynamic control of metabolism case study problem.

	DFBAlab	DC dFBA
Number of elements	–	3
NLP iter	14	97
CPU time (s)	38.72	36.16
t_{reg} (h)	5.42	5.39
Ethanol (mM)	15.06	15.06

3.3. Case study 3: E. coli iJO1366 model

Despite the improvement of the two-stage process in the ethanol titer problem presented in the last section (Section 3.2), the difference between the static strategy and the dynamic control strategy was small. Raj et al. (2020) demonstrated that for the two-stage process to overcome the one-stage process in productivity the second stage must have a non zero growth rate. Moreover, the glucose uptake profile can also have a significant impact on the analysis. Here, we reproduced the results presented in Raj et al. (2020) for the optimization of D-lactic acid productivity in *E. coli* using the DC dFBA. The *E. coli* iJO1366 model was used (1805 metabolites and 2583 reactions) (Orth et al., 2011). The dFBA model used as described by Raj et al. (2020) was:

$$
\begin{aligned}
&\frac{dX}{dt} = \mu X\\
&\frac{dG}{dt} = -v_g X\\
&\frac{dL}{dt} = v_L X\\
&v_g = v_{g,min} + (v_{g,max} - v_{g,min}) * (-1 + \frac{2}{(1 + exp(-K_{up} * \mu))})\\
&X, G, L \geq 0\\
&v_L,\ \mu = pFBA(v_g)
\end{aligned}
\tag{11}
$$

where μ, v_g, v_L are the growth rate, the glucose uptake flux and the D-lactic acid flux, respectively. $K_{up} = 5$ is the uptake constant for the glucose substrate. $v_{g,max} = 10.0$ mmol/gdw h and $v_{g,min} = 0.5$ mmol/gdw h are the maximum and minimum uptake rates for glucose, respectively. The model was solved using the logistic curve for describing the glucose uptake (reduced) as described in Eq. (11) and using a constant glucose uptake rate equal to $v_{g,max}$. Raj et al. (2020) formulated a bi-level optimization problem to maximize the D-lactic acid productivity in a two-stage process. Here, the problem was solved by the DC dFBA in a single-level optimization and the objective function in Eq. (6) was set to $\Phi(x, v) =$ D-lactic acid(end)$/h$. Six finite elements were used with a constraint that the first three elements must have the same growth rate, and the same for the last three elements. Because the size of the elements can change, the NLP can choose the t_{reg} based on the position of these elements.

Fig. 6 show the profiles for both cases (constant and reduced glucose uptake). As already described by Raj et al. (2020), when the glucose uptake is held at its maximum value, there is the first stage of growth alone and a second stage of D-lactic acid production alone. On the other hand, when the effects in the glucose uptake are considered, the best profiles change, and the second optimum stage has a growth rate that decreases the D-lactic acid production. Table 3 shows the change in the stages when glucose uptake is constant or reduced. The maximum growth rate is 0.9 and the maximum v_L is 20. The t_{reg} was similar to the one reported by Raj et al. (2020) demonstrating the capacity of DC dFBA to reproduce the optimal solution using a single-level approach.

3.4. Case study 4: Saccharomyces cerevisiae iND750

Ethanol production by *S. cerevisiae* is still an active subject of research, both in terms of genetic manipulations and bioreactor control and optimization. The dFBA model can be useful in both cases and

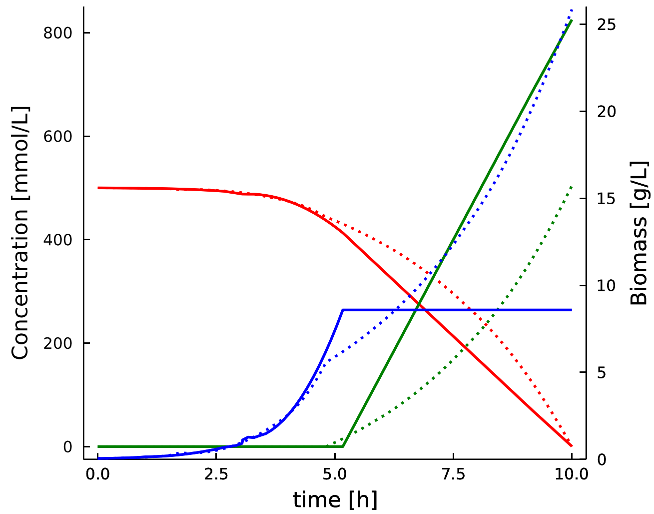

Fig. 6. D-lactic acid production in *E. coli* iJO1366 model for constant glucose uptake (solid line) and reduced glucose uptake (dotted line) using the DC dFBA. The concentration profiles are glucose (red), biomass (blue) and D-lactic acid (green). When the glucose uptake is used the second stage has a null growth rate and the maximum D-lactic acid production. However, when the reduced uptake is applied, the second stage must have a non-zero growth rate.

Table 3
Process parameters for the D-lactic acid production in *E. coli* iJO1366 model using the DC dFBA.

	Glucose uptake	
	Constant	Reduced
μ (1/h)	0.9 ⇒ 0.0	0.9 ⇒ 0.3
v_L (mmol/gdw h)	0.0 ⇒ 20	0.0 ⇒ 7.35
t_{reg} (h)	5.17	4.81
Productivity (mmollac/gdw h)	82.52	50.41
Yield (mmol lac/mmol glu)	1.65	1.01
Titer (mmol lac/gdw)	825	504

the method was applied to simulate the metabolism of *S. cerevisiae* using the GSM iND750 containing 1061 metabolites and 1266 reactions (Duarte et al., 2004). A bioreactor model for the production of ethanol by controlling the glucose feed profile and the dissolved oxygen level was formulated based on previous works (Chang et al., 2016; Oliveira et al., 2021):

$$\begin{aligned}
\frac{dV}{dt} &= F \\
\frac{d(VX)}{dt} &= \mu V X \\
\frac{d(VG)}{dt} &= FG_f - v_g V X \\
\frac{d(VE)}{dt} &= v_e V X \\
v_g &\le v_{g,max} \frac{G}{K_g + G + (G^2/K_{ig})} \frac{1}{1+(E/K_{ie})} \\
v_o &\le v_{o,max} \frac{O}{K_o + O} \\
X, G, E &\ge 0 \\
v_e,\ \mu &= pFBA(v_o, v_g)
\end{aligned} \tag{12}$$

where μ, v_g, v_o, v_e are the growth rate, the glucose uptake flux, oxygen uptake flux and the ethanol flux, respectively. $K_g = 0.5$ g/L, $K_{ig} = 10$ g/L, $K_{ie} = 10.0$ g/L and $K_o = 3.0 \times 10^{-6}$ mol/L are the glucose saturation constant, glucose inhibition constant, ethanol Inhibition constant and oxygen saturation constant, respectively. $v_{g,max} = 20.0$ mmol/gdw h and $v_{o,max} = 8.0$ mmol/gdw h are the maximum uptake rate for glucose and oxygen, respectively. G_f is the feed glucose concentration and was fixed in 50.0 g/L. The control variables were the glucose feed (F) and the oxygen concentration that was assumed to be directly manipulated by a perfect feedback controller. The percent dissolved oxygen is calculated as a function of the oxygen saturation concentration ($DO = O/O_{sat}$) and O_{sat} was set as 3.0×10^{-4} mol/L.

First, a simple simulation was performed in order to compare the performance of the different approaches. Fig. 7 present the results, the RMSE [%] for biomass, glucose, and ethanol for DFBAlab/DA were 0.41/1.75, 0.42/2.35, 0.62/1.44, respectively. The CPU seconds for DC dFBA, DFBAlab, and DA were 203.37, 1.38, 31.85, respectively. As can be seen again, a good agreement between the DC dFBA and dFBAlab was achieved. However, the DA approach failed to represent the profiles after the step in the control variables. The DC dFBA had a considerable increase in time of simulation in comparison with the last examples.

3.4.1. Dynamic optimization of bioreactors

Besides the application of the dFBA model to the dynamic control of the metabolism, the model can also be applied to the dynamic optimization/optimal control of a bioprocess. Chang et al. (2016) formulated a model predictive control using a KKT reformulation of the FBA problem. The authors reported a convergence and initialization problem when they tried to use a GSM and consequently they were forced to apply a core model of *S. cerevisiae*. Oliveira et al. (2021) was able to apply a GSM for the same problem, however, they needed to replace the FBA optimization problem with a surrogate model, adding a training step to the process. Scott et al. (2018) formulated two dynamic optimization problems for *E. coli* and *Pichia stipitis* using GSM. Scott et al. (2018) solved the dFBA as an implicit ODE system and computed the gradient information by solving the sensitivity equation.

The model presented in Eq. (12) was applied to the dynamic optimization of the ethanol production in the bioreactor. The aim is to control the glucose feed flow rate to avoid substrate inhibition and not violate the maximum volume constraint of the bioreactor (1.2 L). The DO inside the reactor must be manipulated to switch from a growth phase (high DO) to a production phase (low DO). The DC dFBA was formulated using the objective function in Eq. (6) as $\Phi(x, v) = ethanol(end)$ and the process constraints were also imposed to the problem ($V_{final} \le 1.2$ and $u_{min} \le u \le u_{max}$). Again, the problem was also solved using DFBAlab + fmincon as a bi-level optimization problem. The control variables were discretized in 10 elements for both approaches.

The results are presented in Fig. 8 and Table 4. Ten different initial guesses were supplied to solve the problem by each method and the solution with the lower objective function (OF) was selected. The DC dFBA reached a solution with a better performance in comparison to the solution found using DFBAlab in terms of final ethanol mass. The DC dFBA profiles have two clearly distinct phases (growth and production) which is analogous to the profiles obtained by Chang et al. (2016) and Oliveira et al. (2021) in recent studies. On the other hand, the DFBAlab profiles were less smooth and because of a higher feed flow at the beginning of the batch, glucose inhibition effects took place. Looking at the numerical aspects, the DC dFBA takes almost the same time to perform a simulation or optimization. This is expected because of the optimization nature of the problem. DFBAlab took a long time to converge using fmincon mostly because of the lack of gradient information, that must be estimated by finite differences, and also because of the presence of nonlinear constraints ($V_{final} \le 1.2$). Another problem with DFBAlab for this application was pointed out by Scott et al. (2018), as the presence of discontinuities in the model at the points where events take place.

It is important to mention that the methodology presented by Scott et al. (2018) represents an alternative for supplying gradient information of dFBA models. Nonetheless, the transcription of the dynamic model into algebraic equation allows the utilization of automatic differentiation packages that makes the solution of optimization problems more straightforward. In contrast, simulation-based computation of

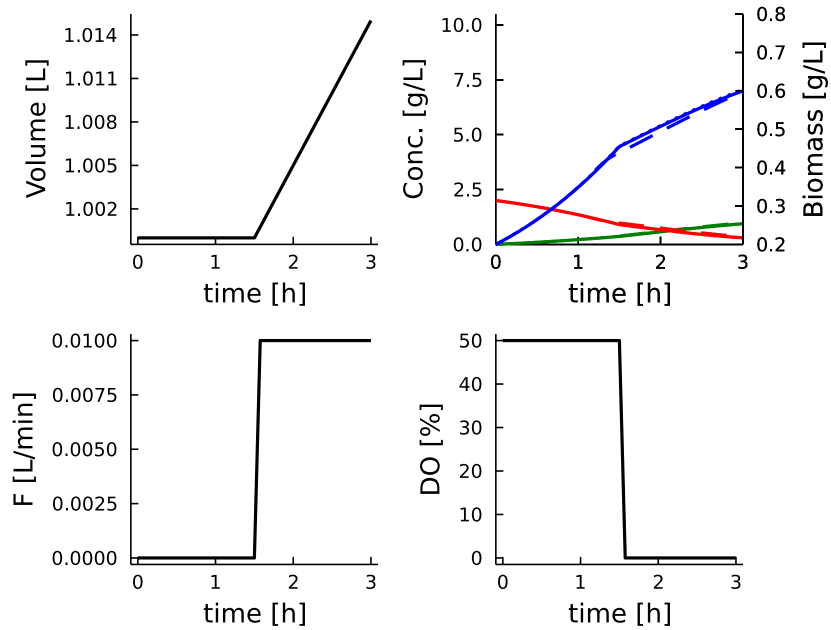

Fig. 7. Bioreactor simulation using the *Saccharomyces cerevisiae* iND750 model. The concentration profiles are glucose (red), biomass (blue) and ethanol (green). DC dFBA (solid), DFBAlab (dotted) and direct approach (dashed). RMSE [%] for biomass, glucose, and ethanol for DFBAlab/DA were 0.41/1.75, 0.42/2.35, 0.62/1.44, respectively. CPU seconds for DC dFBA, DFBAlab and DA were 203.37, 1.38, 31.85, respectively.

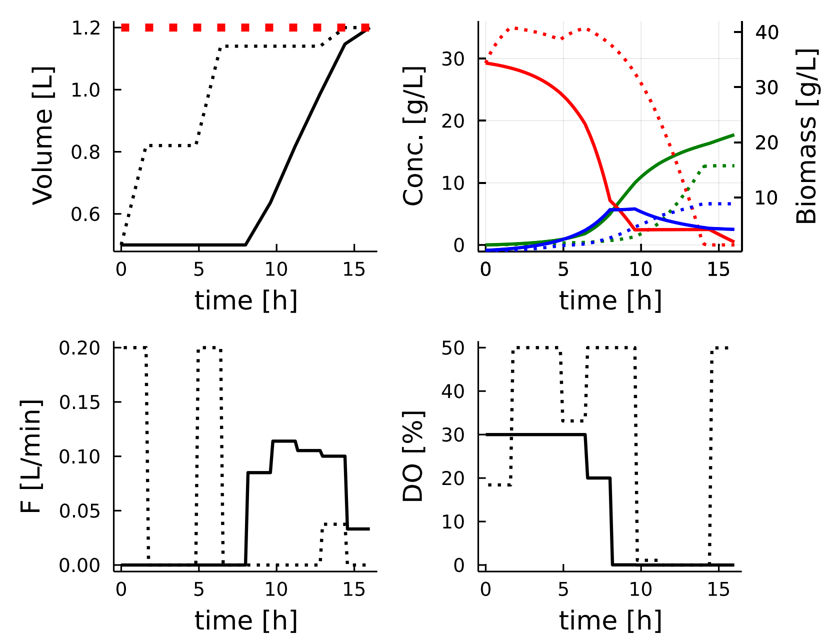

Fig. 8. Dynamic optimization simulation of a bioreactor using the *Saccharomyces cerevisiae* iND750 model using DC dFBA (solid) and DFBAlab (dotted). The concentration profiles are glucose (red), biomass (blue) and ethanol (green).

derivatives information (e.g., finite difference forward sensitivities) are computationally expensive and differential equation solvers can fail at some perturbation inputs making the process unstable (Shin et al., 2019).

3.5. Case study 5: Saccharomyces cerevisiae Yeast 8.3 model

In spite of the fact that dFBA models have fewer parameters than detailed kinetic models, there are still some parameters that need to be estimated using experimental data. As in the dynamic control optimization, the parameter estimation using dFBA becomes a bi-level problem and the non-smoothness of dFBA makes the problem hard to solve. Raghunathan et al. (2006) represent the dFBA model inside the estimation problem using a system of Differential Variational Inequalities that are discretized to yield an MPCC problem and then

Table 4
Comparison of the solution of the dynamic optimization of bioreactor using DC dFBA and DFBAlab.

	DFBAlab	DC dFBA
Number of elements	10	10
Function evaluations	195	268
NLP iterations	6	174
CPU time (min)	40.15	3.27
Ethanol (g)	15.29	21.282

solved using an interior point algorithm. They applied the approach to a small-scale metabolic network of 39 reactions and 43 metabolites. Leppävuori et al. (2011) formulated a sequential gradient-based solution with direct sensitivities equations. They estimated eight parameters and used a metabolic network of 1266 enzymatic reactions and 1061 metabolites. Waldherr (2016) solved a parameter estimation problem for a small-scale network of 10 reactions and 12 metabolites. The bilevel problem was solved as Mixed Integer Quadratic Program. Oliveira et al. (2022) solved a parameter estimation problem using a Yeast metabolic network of 2666 metabolites and 3928 enzymatic reactions. They trained a surrogate model to replace the FBA optimization inside the dFBA model. Five kinetic parameters were estimated: the maximum uptake, saturation constant, and inhibition constant for glucose and xylose uptake.

Here, we applied the DC dFBA to solve the estimation problem formulated by Oliveira et al. (2022) using the Yeast 8.3 metabolic network model (Heavner et al., 2013). The dFBA model describes the *S. cerevisiae* anaerobic growth on glucose and xylose substrates (Eq. (13)).

$$
\begin{aligned}
&\frac{dX}{dt} = \mu X \\
&\frac{dG}{dt} = -v_g X \\
&\frac{dZ}{dt} = -v_z X \\
&\frac{dE}{dt} = v_e X \\
&\quad v_g \le v_{g,max} \frac{G}{K_g + G} \\
&\quad v_z \le v_{z,max} \frac{Z}{K_z + Z} \frac{1}{1 + (G/K_{ig})} \\
&X, G, E, Z \ge 0 \\
&\quad v_e,\ \mu = pFBA(v_g, v_z)
\end{aligned}
\tag{13}
$$

where μ, v_g, v_z, and v_e are the growth rate, and the exchange fluxes of glucose, xylose, and ethanol, respectively. X, G, Z, and E represent the biomass, glucose, xylose, and ethanol concentrations, respectively. $v_{g,max}$ and $v_{z,max}$ are the maximum uptake rate for glucose and xylose respectively. K_g and K_z are the saturation constants, and K_{ie} is the glucose inhibition constant.

A dFBA simulation using the direct method in MATLAB was applied to yield the *in silico* data of glucose, xylose, biomass and ethanol. The set of parameters used is presented in Table 5. The parameter estimation problem was implemented as a DC dFBA using the objective function in Eq. (6) as $\Phi(x, v) = \sum_j (x_j^c(\theta) - x_j^m)^2$. Indexes c and m indicate calculated and measured quantities respectively. Multiple initial guesses were supplied to the solver and the best-fit parameters simulation is presented in Fig. 9. The DC dFBA method was able to find a good fit for the data and the parameters were similar to the ones used to yield the *in silico* data. The features of the performance of DC dFBA can also be found in Table 5. 12 finite elements were used to solve the problem culminating in a large optimization problem to solve. Considering that the problem was solved on a personal laptop, the CPU time was adequate. The uncertainty quantification of the estimated parameters is a crucial step in the model identification process, some techniques rely on solving repeated parameter estimation problems (Shin et al., 2019). Therefore, a fast and reliable method such as DC dFBA is needed.

4. Conclusions

An alternative NLP formulation of dFBA model was presented (DC dFBA) to enable the use of dFBA models in nested optimization problems typically encountered in process optimization and control methods. The method consists of reformulating the pFBA using the KKT conditions and discretizing the dynamic model using the orthogonal collocation technique. The method was implemented in the high-performance JULIA language and solved by the large-scale NLP solver IPOPT. The automatic differentiation code package (Dunning et al., 2017) was applied to supply first and second-order derivative information. The DC dFBA represents a new attempt to solve dFBA models as an optimization problem. As demonstrated, this approach can significantly improve the solution of a class of problems in comparison with the simulation approach. We hope that the tool developed here will contribute to fields like metabolic engineering, synthetic biology, and bioprocess optimization. From the study cases of simulation of *E. coli* diauxic growth on glucose and acetate it was shown that there is a need to impose some flux constraints to guarantee the convergence of the method when there is a change in the active-set of the FBA optimization problem. Furthermore, the advantage of using an adaptive mesh strategy was also highlighted. The DC dFBA was applied to different metabolic networks and scaled well for genome-scale networks. DC dFBA is a much more straightforward way to solve optimization problems with embedded dFBA models. This was demonstrated when the method was applied to simulate dynamic control of metabolism and to solve a dynamic optimization of a bioreactor and a parameter estimation problem.

In future works, some extensions of the methodology and some new applications must be explored, like a quasi-sequential approach that can decrease the CPU time even more (Hong et al., 2006). The methodology developed here could also be extended to problems beyond dFBA

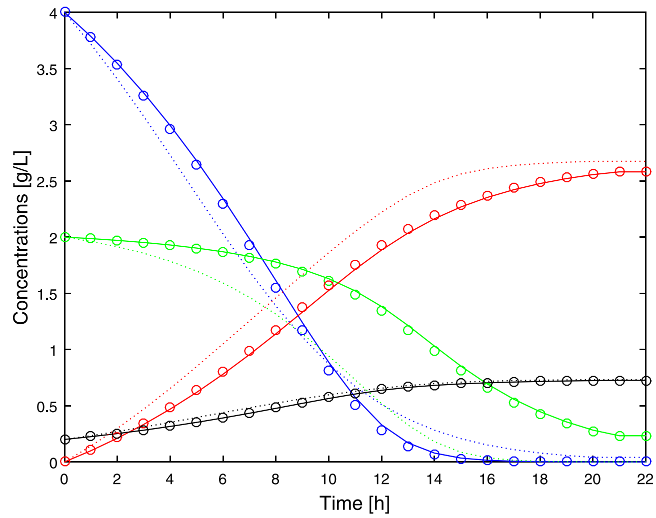

Fig. 9. Best fit of dFBA using *S. cerevisiae* Yeast 8.3 model to the *in silico* data points (circles) computed by DC dFBA (solid line) and the direct approach (dotted line). The concentration profiles are glucose (blue), biomass (black), xylose (green) and ethanol (red).

Table 5
Computational Performance of DC dFBA to solve the parameter estimation problem and the parameter values used in model simulation to yield measurements.

Parameters	Model simulation	DC dFBA	Direct approach
v_g^{max}	7.30	7.11	30.01
K_g	1.03	1.01	12.02
v_z^{max}	32.00	32.88	7.99
K_z	14.85	15.16	0.80
K_{ie}	0.50	0.48	1.00
Number of elements	–	12	–
Variables	–	268 026	–
Constraints	–	362 496	–
CPU	–	26.02 min	27.82 min
Iterations	–	628	5
Function evaluations	–	1440	36
Objective function	–	1.89e−3	7.17

models, such as kinetic models that apply an optimality principle to obtain a solution (Tsiantis et al., 2018).

CRediT authorship contribution statement

Rafael D. de Oliveira: Conceived of the idea, Developed the formulation, Computational framework, Code, Performed the simulations and generated data, Wrote the first draft of the paper. **Galo A.C. Le Roux:** Provided feedback on the idea, Revised the paper, Supervised the project, Provided the funding for this project. **Radhakrishnan Mahadevan:** Conceived of the idea, Revised the paper, Supervised the project, Provided the funding for this project.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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